

SHWETA ARUN SHINDE

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Education

Sardar Patel Institute of Technology, Mumbai

2023 - 2026

Bachelor of Technology in Electronics and Telecommunications, Pursuing 3rd Year, Current CGPA: 7.78

Mumbai

Government Polytechnic Nashik, Nashik

2020 - 2023

Diploma in Electronics and Telecommunications, Percentage: 88.44%

Nashik, Maharashtra

Projects

Automated Dosing Machine for Hydroponic System using IoT and Computer Vision | *AgTech, Embedded System* March

- Developed an automated dosing machine with real-time nutrient optimization, enhancing crop yield through computer vision and precise dosing control.
- Created a user-friendly web interface for remote monitoring and control, improving accessibility and convenience in nutrient management.

PID Controller using FPGA | *Control Systems, VLSI, VHDL*

May 2024

- Developed and implemented a digital PID controller on FPGA using Verilog/VHDL, enhancing control accuracy and real-time response.
- Integrated ADC/DAC for performance validation through simulation and real-time testing.

IoT based Smart Water Heater | *ESP32, Thonny IDE, WiFi, HTML*

October 2024

- Developed a web-based control system for a smart water heater using ESP32, enabling remote monitoring and management of heater settings with real-time environmental data display.
- Implemented multiple control modes and features, including scheduling and energy monitoring, using a digital PID controller for precise temperature management.

Achievements

Winner, Bug Buster 2.0 National Level Hardware Hackathon

February 2025

- Secured 1st place at BugBuster 2.0, a 24-hour national hardware troubleshooting hackathon with 500+ participants, successfully tackling three challenging rounds, including online qualifiers and a rigorous offline round, while showcasing expertise in PCB design, embedded systems, and hardware debugging.

Finalist, IEEE International Future Energy Challenge (IFEC) 2025

November 2024 – Ongoing

- Selected as finalists for the IEEE IFEC 2025 after a successful mid-term review in March 2025 at USA, with our project on Isolated DC-DC Converter.
- Representing India on a global platform in Power Electronics, focusing on energy conversion and sustainable technologies, with the final round scheduled for July 2025.
- Lead the HMI Firmware for the Isolated DC-DC Converter project, ensuring seamless communication and system integration.

VLSI Student Training Program | *Ingot Semi*

June 2024

- Completed a 7-day intensive workshop focused on VLSI design principles, chip design, and semiconductor technology.
- Gained practical exposure to the VLSI design flow, circuit simulations, and layout design techniques.

Technical Skills

Languages: Embedded C, C, C++, Python, Java, HTML, VHDL, Assembly

Developer Tools: VS code, Vivado, Matlab/Simulink, Thonny IDE, Arduino IDE, KiCad, Xilinx, Mide-51, Django, CCStudio, Excel, PowerPoint, Tableau

Development Boards: ESP32, Arduino Uno, Arduino Mega, Rasberry Pi, TMS320F2800137

Technologies: Git, Github

Leadership

Spandan | *Event Coordinator*

February 2023

- Successfully coordinated logistics and communications, enhancing event efficiency and participant engagement.

TechVolts | *Operation Executive*

September 2022

- Led the logistics for a Treasure Hunt event involving 30 teams, fostering collaboration and problem-solving skills.